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Consultatie 21.01.

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Integrale de suprafață

1) Parametrizați următoarele suprafețe:

a) $x^2+y^2=4$ → proiecția pe xOy este un cerc

2 variabile ⇒ cilindru

$$\begin{cases} x = 2\cos u \\ y = 2\sin u \\ z = v \end{cases}$$

Parametrizarea
$u \in [0, 2\pi)$
$v \in \mathbb{R}$

$$(u,v) \xrightarrow{\varphi(u,v)} \begin{cases} x = X(u,v) \\ y = Y(u,v) \\ z = Z(u,v) \end{cases}$$

b) $z = x^2+y^2+1$

↳ paraboloid

$$\begin{cases} x = x \\ y = y \\ z = x^2+y^2+1 \end{cases}$$

Parametrizarea carteziană

$(x,y) \in \mathbb{R}^2$

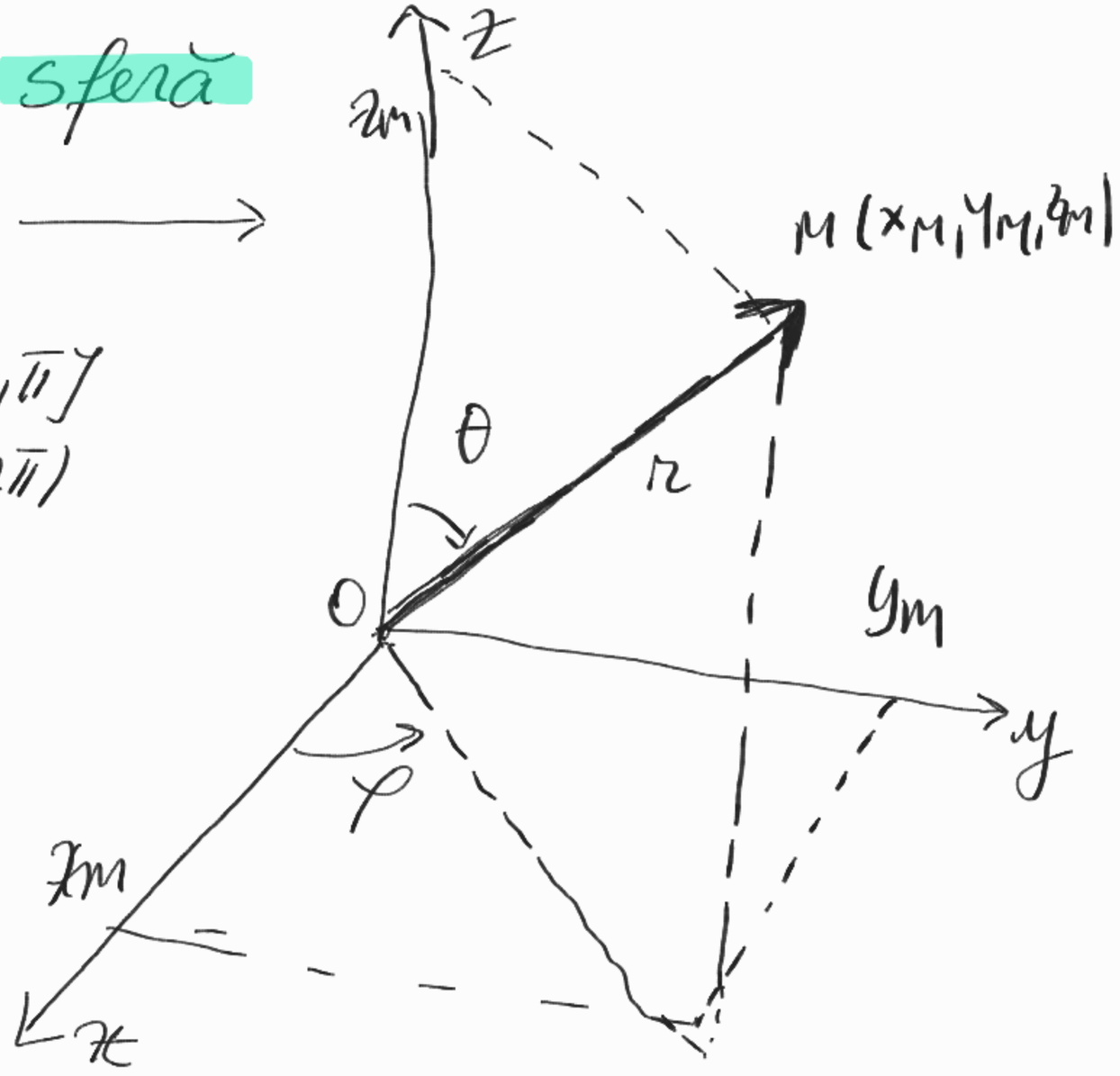
$$\begin{cases} x = u\cos v \\ y = u\sin v \\ z = u^2+1 \end{cases} \qquad \begin{matrix} v \in [0, 2\pi) \\ u \in [0, +\infty) \end{matrix}$$

↓ coordonate polare

c) $x^2+y^2+z^2=16$ → sferă

Coordonate sferice →

$$\begin{cases} x = 4\sin\theta\cos\varphi & \theta\in[0,\pi] \\ y = 4\sin\theta\sin\varphi & \varphi\in[0,2\pi) \\ z = 4\cos\theta \end{cases}$$



d) $x^2+2y^2+3z^2=4 \Leftrightarrow \frac{x^2}{4}+\frac{y^2}{2}+\frac{z^2}{4/3}=1$

↓ elipsoid

$$\frac{x^2}{2^2}+\frac{y^2}{(\sqrt{2})^2}+\frac{z^2}{(2/\sqrt{3})^2}=1$$

$$\Rightarrow \begin{cases} \frac{x}{2} = \sin\theta\cos\varphi \\ \frac{y}{\sqrt{2}} = \sin\theta\sin\varphi \\ \frac{z}{2/\sqrt{3}} = \cos\theta \end{cases} \Rightarrow \begin{cases} x = 2\sin\theta\cos\varphi & \theta\in[0,\pi] \\ y = \sqrt{2}\sin\theta\sin\varphi & \varphi\in[0,2\pi] \\ z = \frac{2}{\sqrt{3}}\cos\theta \end{cases}$$

e) $z=\sqrt{x^2+y^2}$, $z\in[0,2]$ → con

$$\begin{cases} x = x \\ y = y \\ z = \sqrt{x^2+y^2} \end{cases}$$ Param. carteziană

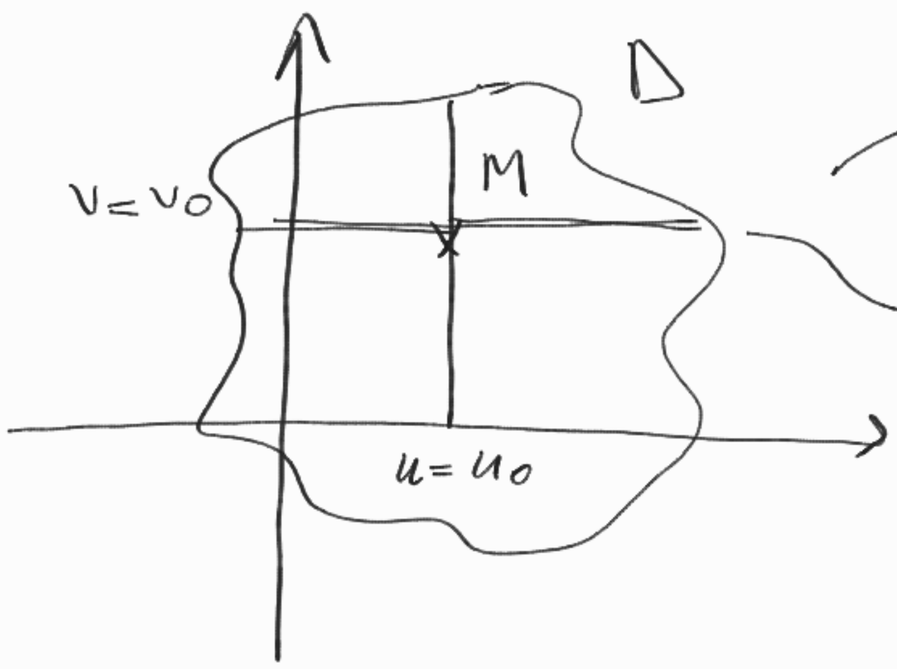
$z\in[0,2]$

$(x,y)\in\{(x,y)\in\mathbb{R}^2 \mid x^2+y^2\le 4\} = D$ (disc de rază 2)

$$\begin{cases} x = u\cos v & u\in[0,2] \\ y = u\sin v & v\in[0,2\pi] \\ z = \sqrt{u^2} = u \end{cases}$$

(2) Aflați aria suprafeței conului $z = \sqrt{x^2+y^2}$ ce se găsește în int. cilindrului $x^2+y^2 = 2x$

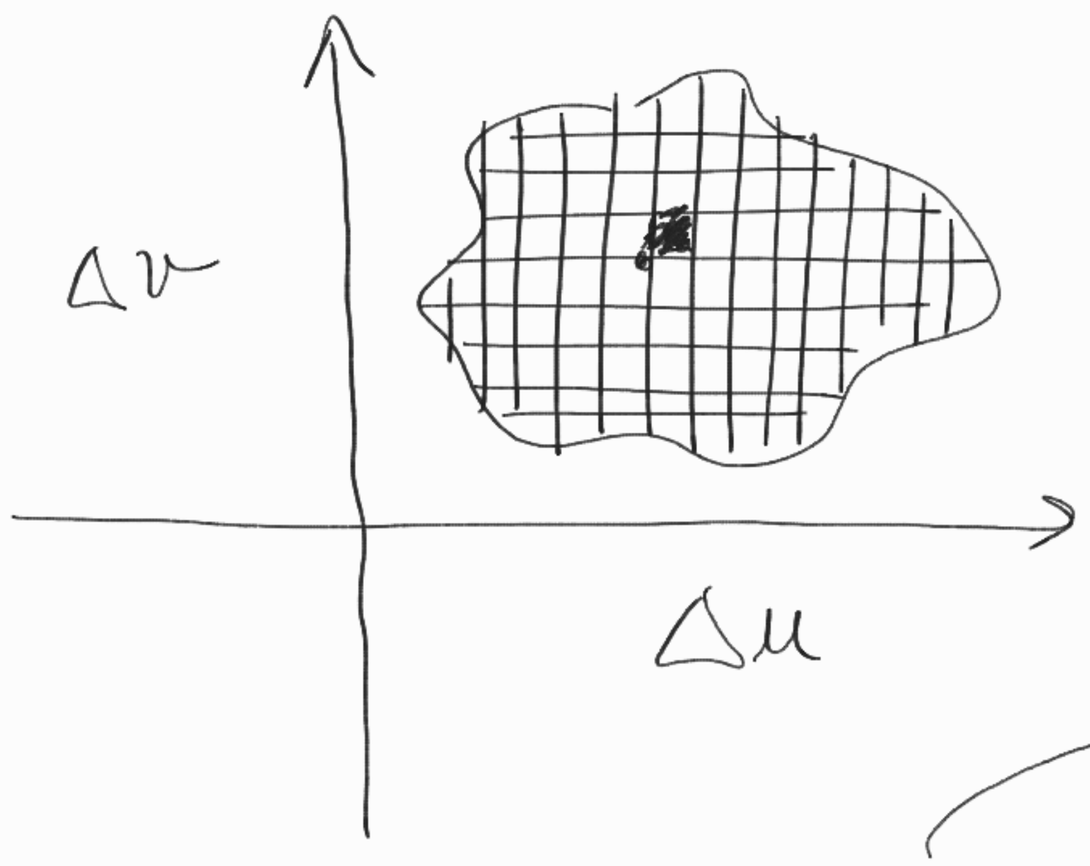
Teorie → vezi Pdf Negrescu



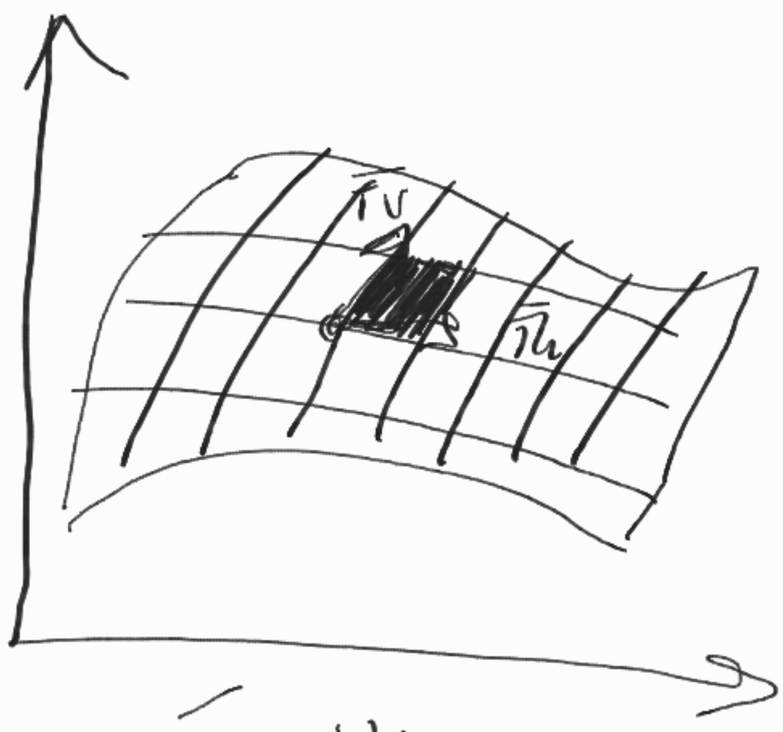
φ

→ imaginea segmentului este o curbă

(u,v) — D — φ → $\begin{cases} x = X(u,v) \\ y = Y(u,v) \\ z = Z(u,v) \end{cases}$ — S



φ



Aria suprafeței = ?

" suma ariilor

" peticelor "

$$\boxed{A(S) = \sum\sum A(\text{petic})} = \text{Aria paralelogramului}$$

↳ calcul pe două dimensiuni

$$= \sum\sum \| \bar{T}_u \Delta u \times \bar{T}_v \Delta v \| = \sum\sum \| \bar{T}_u \times \bar{T}_v \| \cdot \Delta u \cdot \Delta v$$

Trecem la limită $\xrightarrow{\lim}$ $\lim \left(\sum\sum \| \bar{T}_u \times \bar{T}_v \| \Delta u \Delta v \right)$

$\rightarrow \iint_D \| \bar{T}_u \times \bar{T}_v \| \, du\,dv \rightarrow$ Aria unei suprafețe $\Delta\sigma$

$\| \bar{T}_u \times \bar{T}_v \| = A(S)$

$\hookrightarrow \iint_S d\sigma$ (Notație)

$$\bar{T}_u = \frac{\partial x}{\partial u}\bar{i} + \frac{\partial y}{\partial u}\bar{j} + \frac{\partial z}{\partial u}\bar{k}$$

$$\bar{T}_v = \frac{\partial x}{\partial v}\bar{i} + \frac{\partial y}{\partial v}\bar{j} + \frac{\partial z}{\partial v}\bar{k}$$

> Vectori tangenți

Deci aria unei suprafețe este $\iint_S d\sigma = \iint_D \| \bar{T}_u \times \bar{T}_v \| \, du\,dv$

unde $\bar{T}_u$, $\bar{T}_v$ au formulele.

$\iint_S$ → integrală de suprafață

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(2) Aria suprafeței delimitate de:

$\begin{cases} \text{conul } z = \sqrt{x^2+y^2} \\ \text{cilindrul } x^2+y^2 = 2x \Rightarrow (x-1)^2+y^2 = 1 \end{cases}$

$\begin{cases} x = x \\ y = y \\ z = \sqrt{x^2+y^2} \end{cases}$ Parametrizarea conului

("petala" este pe con)

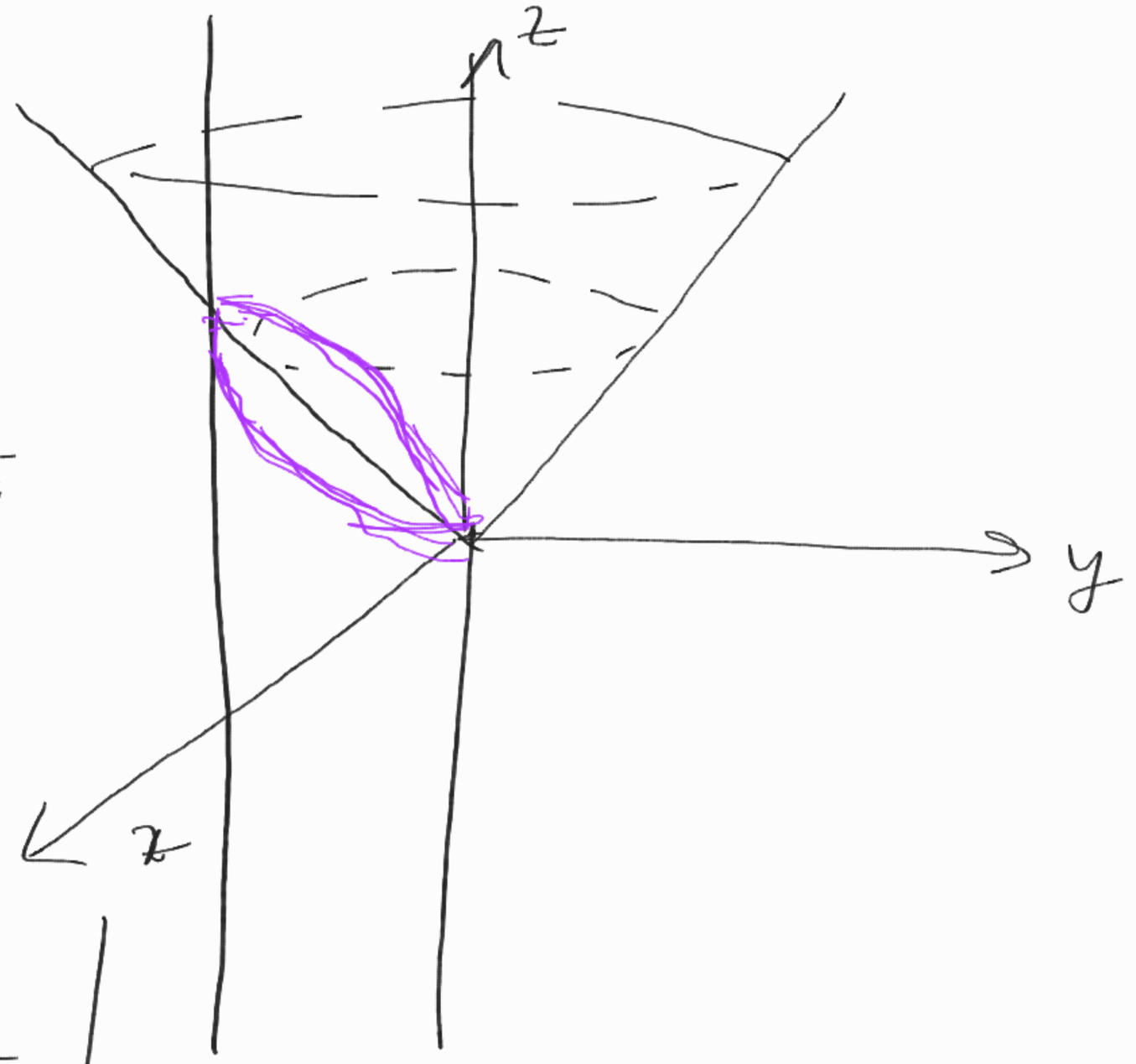
$D: x^2+y^2 \le 2x$ Determinăm integrala de suprafață.

$$\bar{T}_x = 1\cdot\bar{i} + 0\cdot\bar{j} + \frac{2x}{2\sqrt{x^2+y^2}}\cdot\bar{k}$$

$$= \bar{i} + \frac{x}{\sqrt{x^2+y^2}}\bar{k}$$

$$\bar{T}_y = 0\cdot\bar{i} + 1\cdot\bar{j} + \frac{2y}{2\sqrt{x^2+y^2}}\cdot\bar{k}$$

$$= \bar{j} + \frac{y}{\sqrt{x^2+y^2}}\cdot\bar{k}$$



$$\bar{T}_x \times \bar{T}_y = \begin{vmatrix} \bar{i} & \bar{j} & \bar{k} \\ 1 & 0 & \frac{x}{\sqrt{x^2+y^2}} \\ 0 & 1 & \frac{y}{\sqrt{x^2+y^2}} \end{vmatrix}$$

$$= \frac{-x}{\sqrt{x^2+y^2}}\bar{i} - \frac{y}{\sqrt{x^2+y^2}}\bar{j} + \bar{k}$$

$$\| \bar{T}_x \times \bar{T}_y \| = \sqrt{\frac{x^2}{x^2+y^2} + \frac{y^2}{x^2+y^2} + 1} = \sqrt{2}$$

$$\Rightarrow A(S) = \iint_D \| \bar{T}_x \times \bar{T}_y \| \, dx\,dy = \iint_D \sqrt{2}\, dx\,dy =$$

$\sqrt{2}\cdot$ Arie

$$\sqrt{2}\iint_D dx\,dy = \sqrt{2}\int_{-\pi/2}^{\pi/2}\int_0^{2\cos t} r\, dr\, dt = \sqrt{2}\int_{-\pi/2}^{\pi/2} \frac{r^2}{2}\Big|_0^{2\cos t} dt =$$

$$= 2\sqrt{2}\int_{-\pi/2}^{\pi/2} \underbrace{\cos^2 t}_{\frac{1+\cos 2t}{2}}\, dt =$$

$$= 2\sqrt{2}\left(\frac{t}{2} + \frac{\sin 2t}{4}\right)\Big|_{-\frac{\pi}{2}}^{\frac{\pi}{2}} = \pi\sqrt{2}$$

Met 1

$D: x^2+y^2 \le 2x$

$r^2 \le 2r\cos t$

$r \le 2\cos t$

$r \in [0, 2\cos t]$

$t \in [-\pi/2, \pi/2]$

(3.) Fereastra lui Viviani

Aria suprafeței de pe sfera $x^2+y^2+z^2=1$ intersectată cu cilindrul $x^2+y^2=y$ pentru $z\geq 0$.

Soluție

$$\begin{cases} x^2+y^2+z^2=1 \\ x^2+y^2=y \\ z\geq 0 \end{cases}$$

Bucata se află pe sferă, deci parametrizăm sfera.

$$\begin{cases} x=x \\ y=y \\ z=\sqrt{1-x^2-y^2} \end{cases} \rightarrow \text{parametrizare carteziană}$$

D: $x^2+y^2\leq y$ (interiorul cilindrului)

$$\bar{T_x}=\bar{i}-\frac{x}{\sqrt{1-x^2-y^2}}\bar{k}$$

$$\bar{T_y}=\bar{j}-\frac{y}{\sqrt{1-x^2-y^2}}\bar{k}$$

$$\Rightarrow \bar{T_x}\times\bar{T_y}=\begin{vmatrix} \bar{i} & \bar{j} & \bar{k} \\ 1 & 0 & \frac{-x}{\sqrt{1-x^2-y^2}} \\ 0 & 1 & \frac{-y}{\sqrt{1-x^2-y^2}} \end{vmatrix}=\frac{x}{\sqrt{1-x^2-y^2}}\bar{i}+\frac{y}{\sqrt{1-x^2-y^2}}\bar{j}+\bar{k} \Rightarrow$$

$$\|\bar{T_x}\times\bar{T_y}\|=\sqrt{\frac{x^2}{1-x^2-y^2}+\frac{y^2}{1-x^2-y^2}+1}=\frac{1}{\sqrt{1-x^2-y^2}}$$

$$\Rightarrow \text{Aria}(S)=\iint_D \frac{dx\,dy}{\sqrt{1-x^2-y^2}} \overset{*}{=}$$

Trecem la coordonate polare.

$r^2 \le r\sin t \Rightarrow r \le \sin t \Rightarrow \begin{cases} r \in [0, \sin t] \\ t \in [0, \pi] \end{cases}$

$\overset{*}{=} -\frac{1}{2}\int_0^{\pi}\int_0^{\sin t} \frac{-2r}{\sqrt{1-r^2}} \xrightarrow{\text{jacobian}} dr\,dt = -\frac{1}{2}\int_0^{\pi} \frac{\sqrt{1-r^2}}{1/2}\Big|_0^{\sin t} dt =$

$= -\int_0^{\pi} (|\cos t| - 1)\, dt = -\int_0^{\pi/2} (\cos t - 1)\, dt + \int_{\pi/2}^{\pi} (-\cos t - 1)\, dt$

$= \pi - 2$

Integrale de suprafață de speța I

④ $\iint_S x^2 y^2 z \, d\sigma = ?$, unde S este porțiunea conului de ecuație $z = \sqrt{x^2+y^2}$, $z \in [0;1]$

Soluție

$\iint_S \underbrace{x^2 y^2 z}_{f(x,y,z)} \, d\sigma \qquad S: z = \sqrt{x^2+y^2}, z \in [0;1]$

$\iint_S f(x,y,z)\, d\sigma = \iint_D f(x(u,v), y(u,v), z(u,v)) \cdot \|\bar{T}_u \times \bar{T}_v\| \, du\, dv$

↳ formula

$$\begin{cases} x = R\cos t & R \in [0,1] \\ y = R\sin t & t \in [0, 2\pi] \\ z = R \end{cases}$$

$$\bar{T}_R = \cos t\,\bar{i} + \sin t\,\bar{j} + \bar{k}$$

$$\bar{T}_t = -R\sin t\,\bar{i} + \cos t\,R\,\bar{j}$$

$$\bar{T}_R \times \bar{T}_t = \begin{vmatrix} \bar{i} & \bar{j} & \bar{k} \\ \cos t & \sin t & 1 \\ -R\sin t & R\cos t & 0 \end{vmatrix} = -R\cos t\,\bar{i} - R\sin t\,\bar{j} + R\bar{k}$$

$$\Rightarrow \| \bar{T}_R \times \bar{T}_t \| = \sqrt{R^2\cos^2 t + R^2\sin^2 t + R^2} = R\sqrt{2}$$

$$\Rightarrow \iiint_S f(x,y,z)\,dv = \int_0^{2\pi}\int_0^1 R^5 \sin^2 t \cos^2 t\, R\sqrt{2}\,dr\,dt =$$

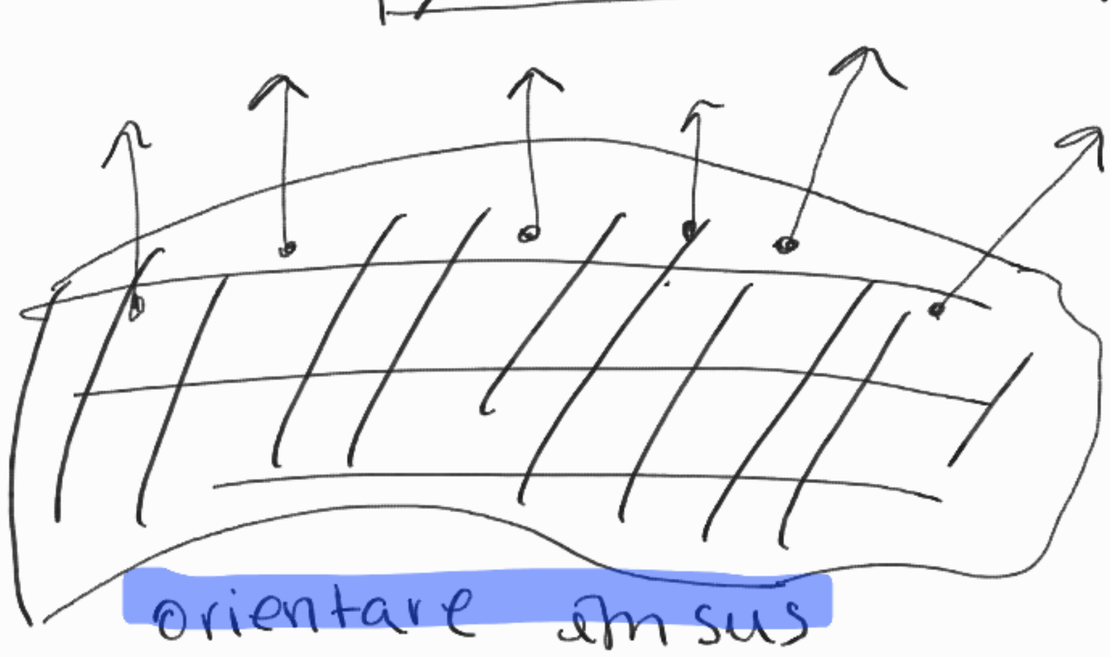
$$\sqrt{2}\int_0^1 R^6\,dr \cdot \int_0^{2\pi} \sin^2 t\cos^2 t\,dt = \frac{\sqrt{2}}{7}\cdot\left(\frac{t}{8} - \frac{\sin 4t}{32}\right)_0^{2\pi} = \frac{\pi\sqrt{2}}{28}$$

$$\sin^2 2t = \frac{1-\cos 4t}{2}$$

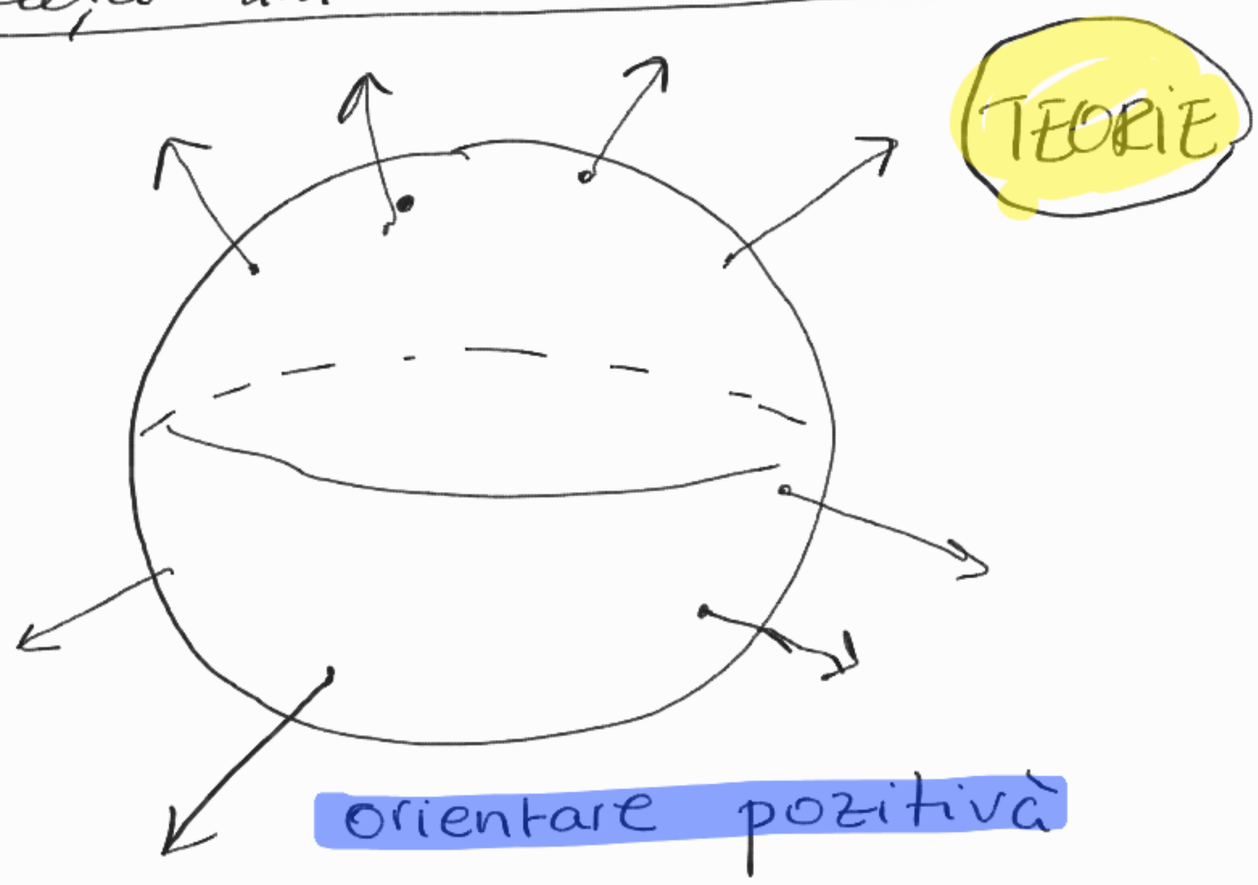
Fluxul

unui câmp vectorial

↓ **cantitatea de volum care trece printr-o suprafață într-o unitate de timp**

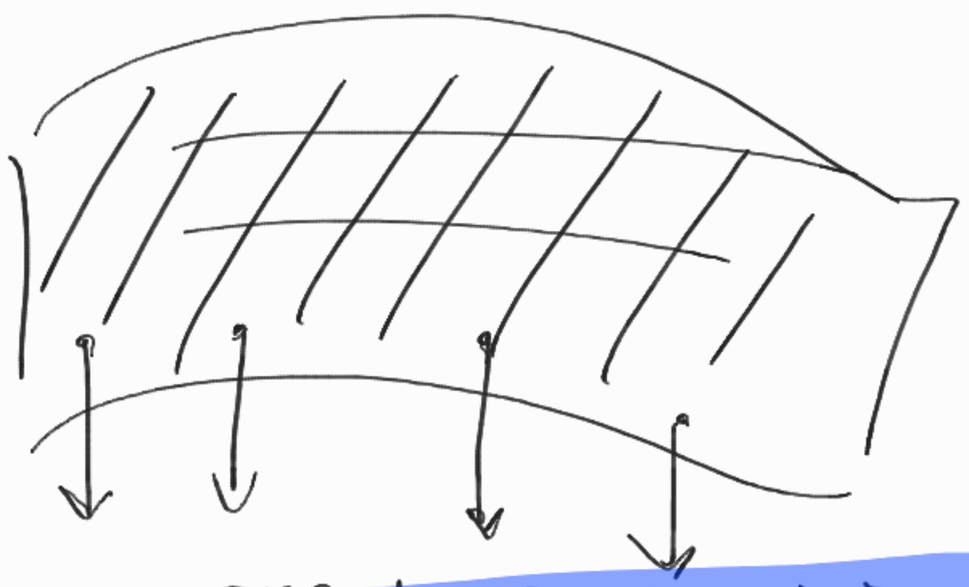


orientare în sus
(↑ crescător)

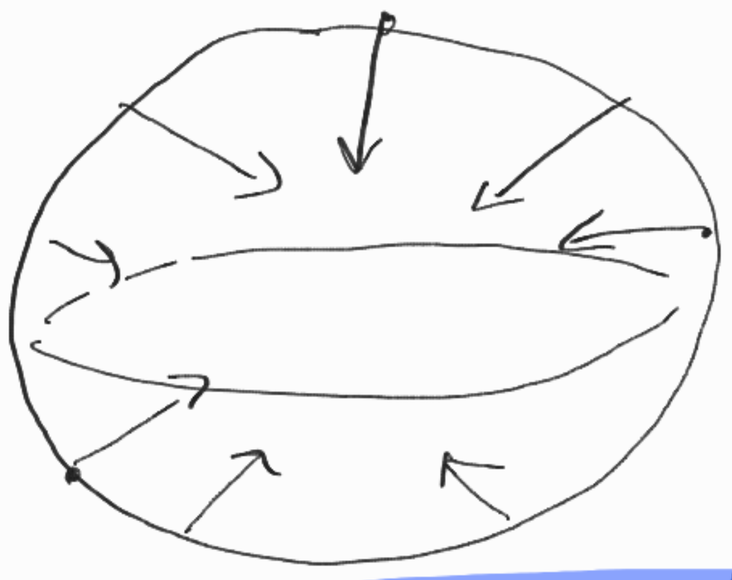


TEORIE

orientare pozitivă

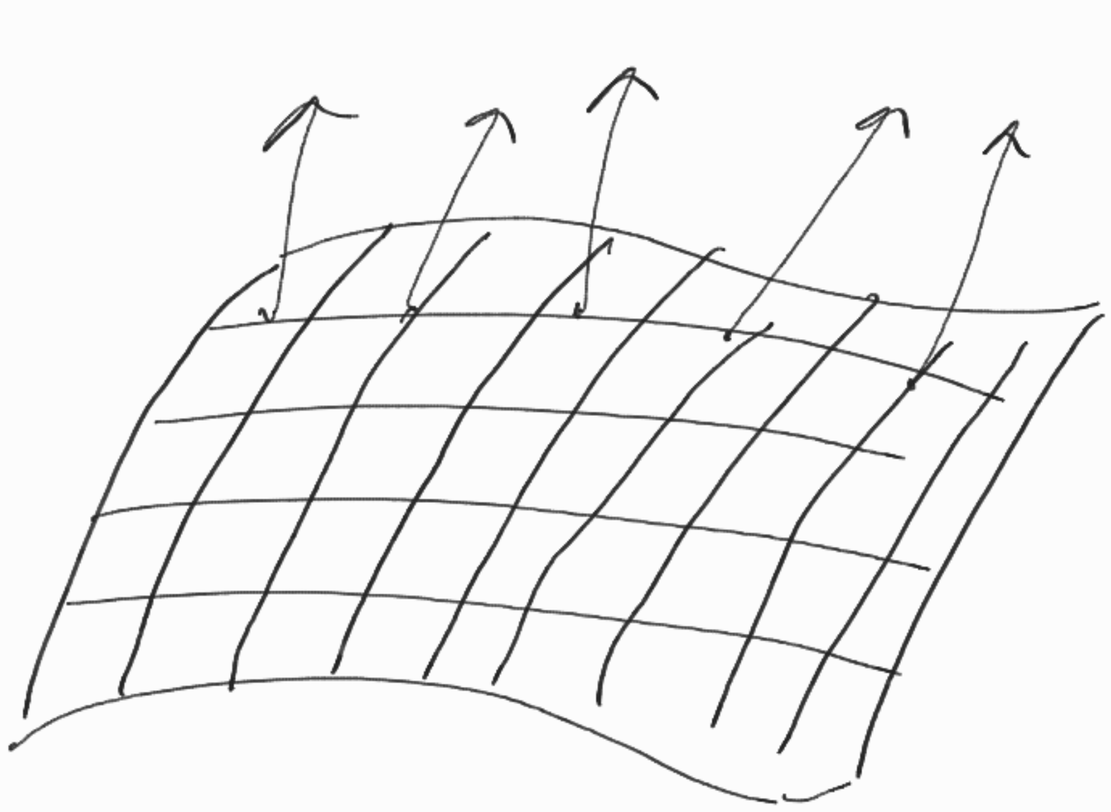


orientare în jos
(↓ descrescător)



orientare negativă

Există și suprafețe care nu au 2 orientări
(Banda lui Möbius)



$\bar{n} \rightarrow$ versor

$$pr_{\bar{n}}\bar{F} = \frac{\langle \bar{F}, \bar{n} \rangle}{\langle \bar{n}, \bar{n} \rangle} \cdot \bar{n} =$$

$$= \langle F, n \rangle \cdot n$$

$\| pr_{\bar{n}}\bar{F} \| = \bar{F} \cdot \bar{n}$ ⟶ altă notație

$d = \bar{F} \cdot \bar{n}$

$\Rightarrow$ volumul este $\boxed{\bar{F} \cdot \bar{n} \cdot \Delta\bar{V}} = F$ (flux)

$\Rightarrow$ $\mathcal{F}_S(\bar{F}) = \boxed{\lim \sum \sum} \bar{F} \cdot \bar{n} \, \Delta\bar{V} = \iint_S \bar{F} \cdot \bar{n} \, d\bar{V}$

↓ FORMULA FLUXULUI

$$\mathcal{F}_S(\bar{F}) = \iint_S \bar{F} \cdot \bar{n} \, d\bar{V} = \iint_D \bar{F} \cdot \frac{\bar{T}_u \times \bar{T}_v}{\| \bar{T}_u \times \bar{T}_v \|} \| \bar{T}_u \times \bar{T}_v \|$$

$$du\,dv = \iint_D \bar{F} \cdot (\bar{T}_u \times \bar{T}_v) \, du\,dv$$

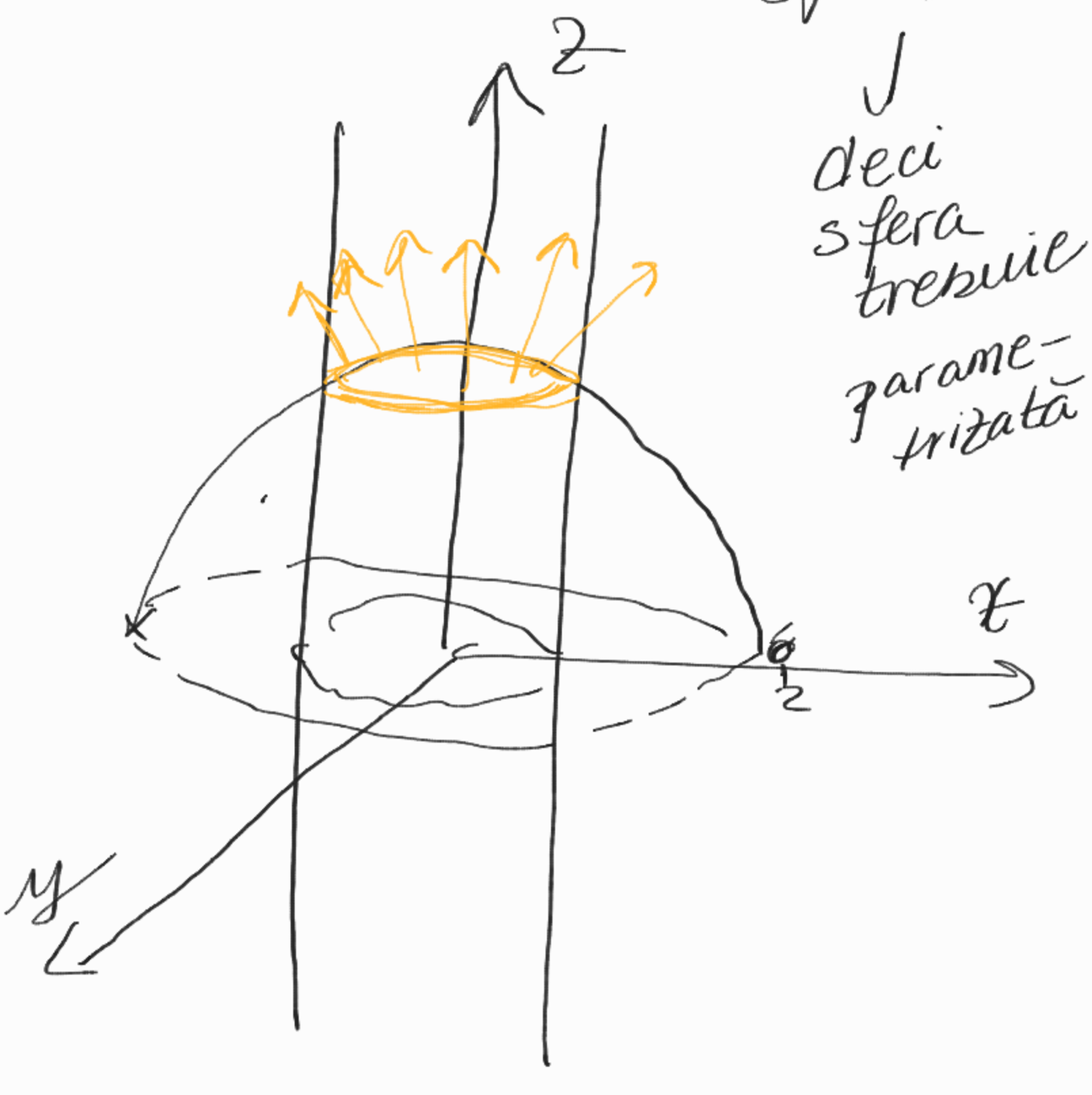
produs mixt

produs scalar

produs vectorial

5. $F(x,y,z) = \frac{x}{3}\bar{i} + \frac{y}{3}\bar{j} + \frac{z}{3}\bar{k}$ fluxul prin suprafața sferei

$\begin{cases} x^2+y^2+z^2=4 \\ x^2+y^2 \le 1 \end{cases}$ sferă $z \ge 0$



↓ deci sfera trebuie parametrizată

$x = x$

$y = y$

$z = \sqrt{4-x^2-y^2}$

$D: x^2+y^2 \le 1$

$\bar{T_x} = \bar{i} - \frac{x}{\sqrt{4-x^2-y^2}}\bar{k}$

$\bar{T_y} = \bar{j} - \frac{y}{\sqrt{4-x^2-y^2}}\bar{k}$

$\Rightarrow \bar{T_x} \times \bar{T_y} =$

$= \begin{vmatrix} \bar{i} & \bar{j} & \bar{k} \\ 1 & 0 & \frac{-x}{\sqrt{4-x^2-y^2}} \\ 0 & 1 & \frac{-y}{\sqrt{4-x^2-y^2}} \end{vmatrix}$

$= \frac{x}{\sqrt{4-x^2-y^2}}\bar{i} + \frac{y}{\sqrt{4-x^2-y^2}}\bar{j} + \bar{k}$

Dacă obținem a treia comp. de semn opus rezultatul este $-\not\exists(S)$

↓ orientare pozitivă (în sus) !!!

de la parametrizare

$$\Phi_S(\vec{F}) = \iint_D \left(\frac{x}{3}, \frac{y}{3}, \frac{\sqrt{4-x^2-y^2}}{3}\right) \left(\frac{x}{\sqrt{4-x^2-y^2}}, \frac{y}{\sqrt{4-x^2-y^2}}, 1\right)$$

$$dx\,dy = \iint_D \left(\frac{x^2+y^2}{3\sqrt{4-x^2-y^2}} + \frac{\sqrt{4-x^2-y^2}}{3}\right) dx\,dy =$$

$$= \iint_D \frac{4}{3\sqrt{4-x^2-y^2}}\, dx\,dy = *$$

Trecem la coordonate polare $R \in [0,1]$, $t \in [0, 2\pi]$

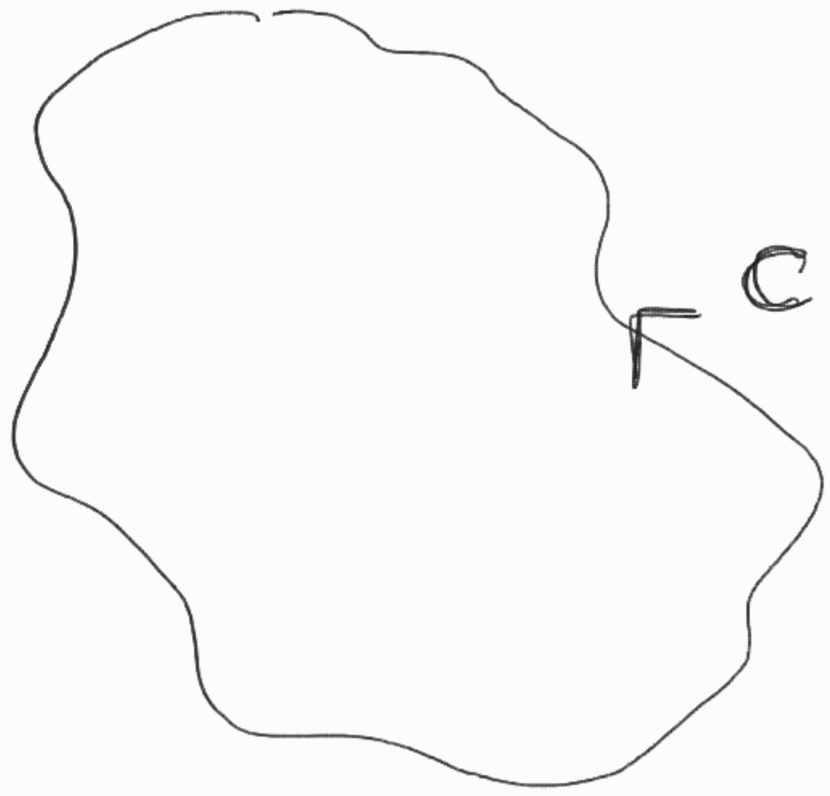
$$* = \int_0^{2\pi}\int_0^1 \frac{R}{\sqrt{4-R^2}}\, dR\,dt = \frac{4}{3}\int_0^{2\pi} dt \cdot \int_0^1 \frac{R\,dR}{\sqrt{4-R^2}} =$$

$\rightarrow$ jacobianul

= - - - - - calculele

FORMULA G-R

(Green - Riemann)



$P, Q \in C^1$ pe o mulțime ce conține curba C

$$\int_C P\,dx + Q\,dy = \iint_D \left(\frac{\partial Q}{\partial x} - \frac{\partial P}{\partial y}\right) dx\,dy$$

Problemă

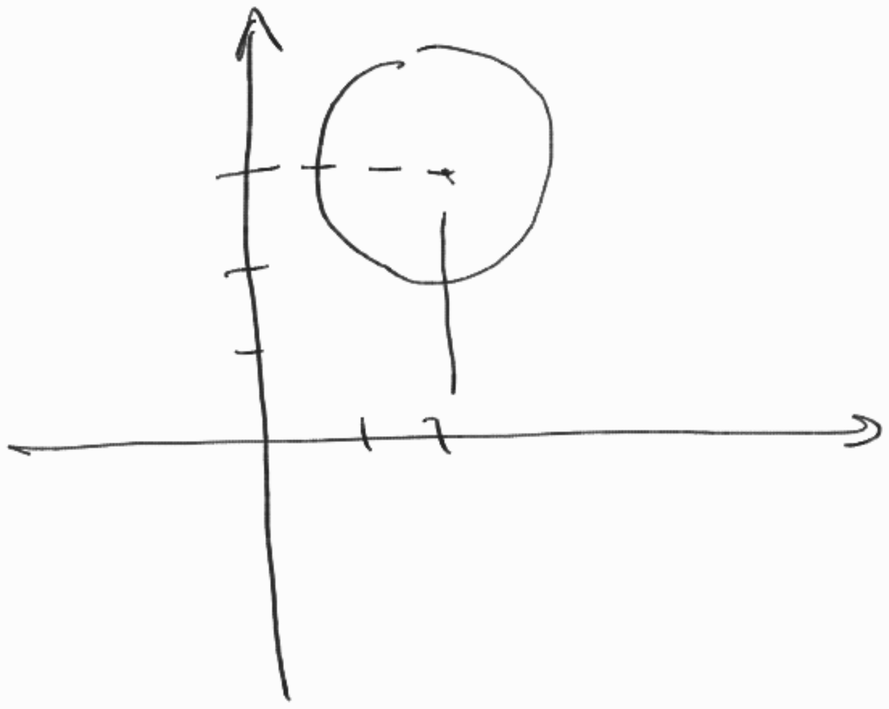
$$\int_C \underbrace{\sqrt{x^2+y^2}}_{P}\,dx + \underbrace{y\left[4y + \ln\left(x+\sqrt{x^2+y^2}\right)\right]}_{Q}\,dy$$

$$C = \{(x,y) \in \mathbb{R} \mid x^2+y^2-4x-6y+12=0\}$$

Soluție

$$C: x^2+y^2-4x-6y+12=0 \iff (x-2)^2+(y-3)^2=1$$

↓ cerc de centru $(2,3)$ și rază 1



Calculăm $\frac{\partial Q}{\partial x} - \frac{\partial P}{\partial y} = y\left(y + \frac{1}{x+\sqrt{x^2+y^2}} \cdot \left(1 + \frac{x}{\sqrt{x^2+y^2}}\right)\right) - \frac{y}{\sqrt{x^2+y^2}} = y^2 + y\frac{1}{x+\sqrt{x^2+y^2}} \cdot \frac{x+\sqrt{x^2+y^2}}{\sqrt{x^2+y^2}} - \frac{y}{\sqrt{x^2+y^2}} = y^2 \Rightarrow$

$$\int_C P\,dx + Q\,dy = \iint_D y^2\,dx\,dy = \int_0^{2\pi}\int_0^1 (3+R\sin t)^2 \cdot R\,dR\,dt$$

$D: (x-2)^2 + (y-3)^2 \le 1$

$$\begin{cases} x = R\cos t + 2 \\ y = R\sin t + 3 \end{cases} \qquad R^2 \le 1 \Rightarrow R \in [0,1], \; t \in [0, 2\pi]$$

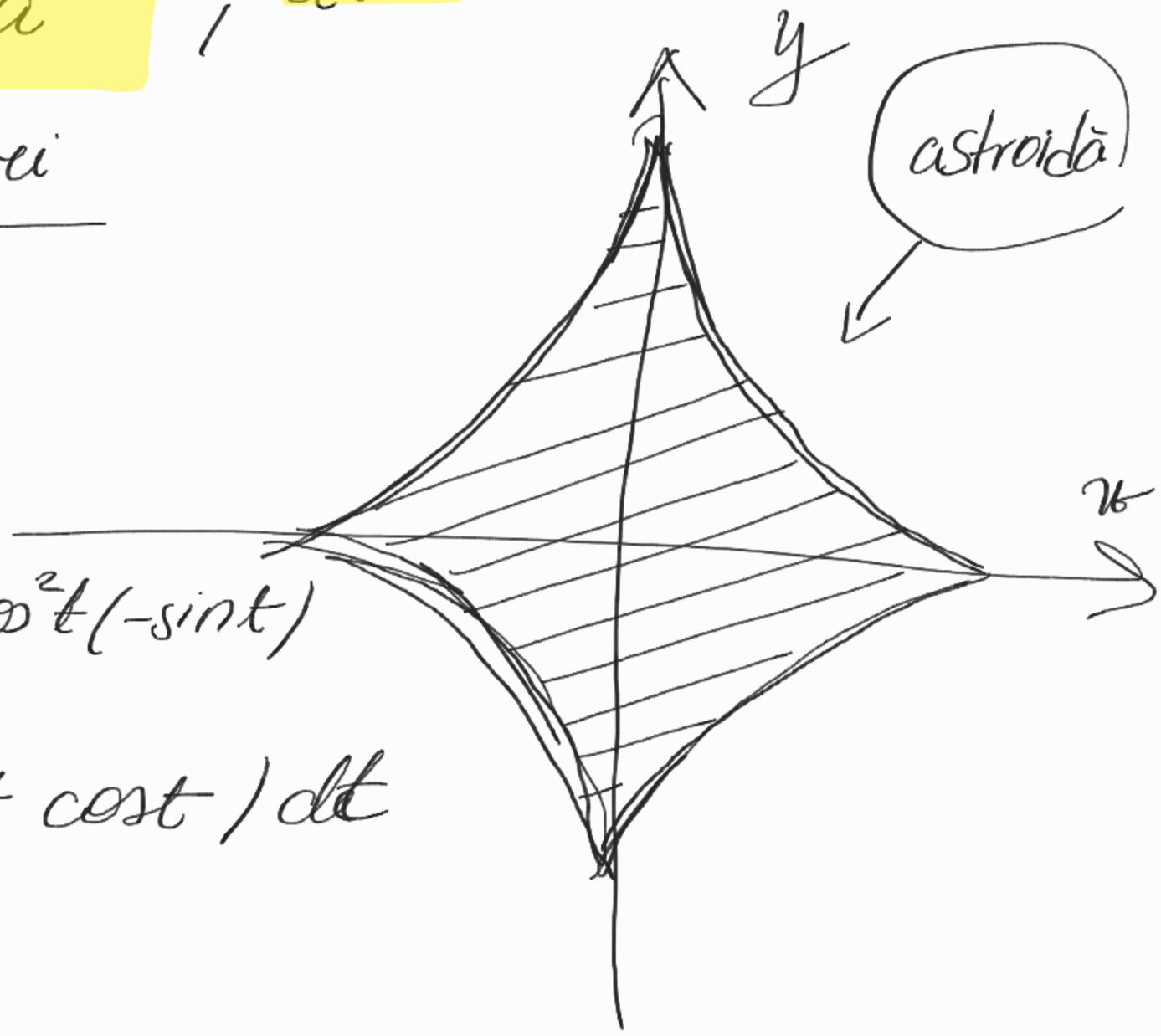
$$= \int_0^{2\pi}\int_0^1 (9R + 6R^2\sin t + R^3\sin^2 t)\,dR\,dt = \text{(calculează)} \; \dots$$

$$A(D) = \iint_D 1\,dx\,dy = \int_C \underbrace{-\frac{y}{2}}_{P}\,dx + \underbrace{\frac{x}{2}}_{Q}\,dy$$

$\uparrow$ $\frac{\partial Q}{\partial x} - \frac{\partial P}{\partial y}$

(8) Aria sup. mărg. de astroida

$$x^{2/3} + y^{2/3} = a^{2/3}, \quad a > 0$$



Parametrizarea curbei

$$C: \begin{cases} x = a\cos^3 t \\ y = a\sin^3 t \end{cases} \quad t \in [0, 2\pi]$$

$$A(S) = \frac{1}{2}\int_0^{2\pi} (-a\sin^3 t \cdot 3a\cos^2 t(-\sin t) + a\cos^3 t \cdot 3a\sin^2 t \cos t)\, dt$$

$$= \dots$$

(9) Forma diferențială

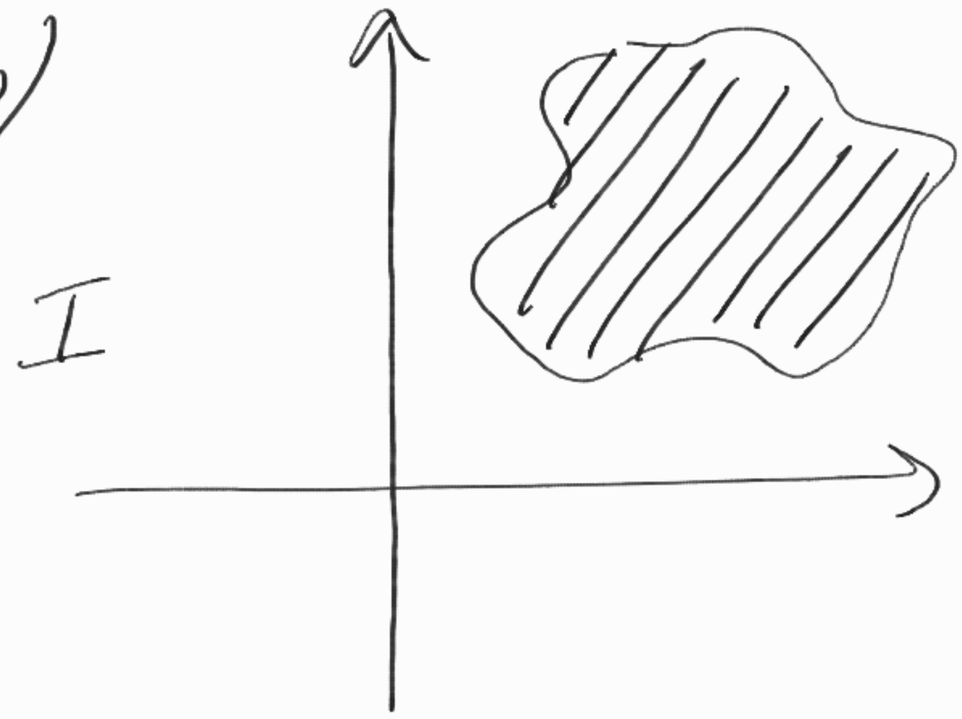
$$w = \frac{-y}{x^2+y^2}\, dx + \frac{x}{x^2+y^2}\, dy$$

a) $\int_{\partial b(0,R)} (w) = \int_{b(0,R)} \underbrace{\frac{-y}{x^2+y^2}\, dx + \frac{x}{x^2+y^2}\, dy}_{w}$

$$= \int_0^{2\pi} \left(-\frac{R\sin t}{R^2}(-R\sin t) + \frac{R\cos t}{R^2} \cdot R\cos t\right) dt =$$

$$= \int_0^{2\pi} 1\, dt = 2\pi$$

b)



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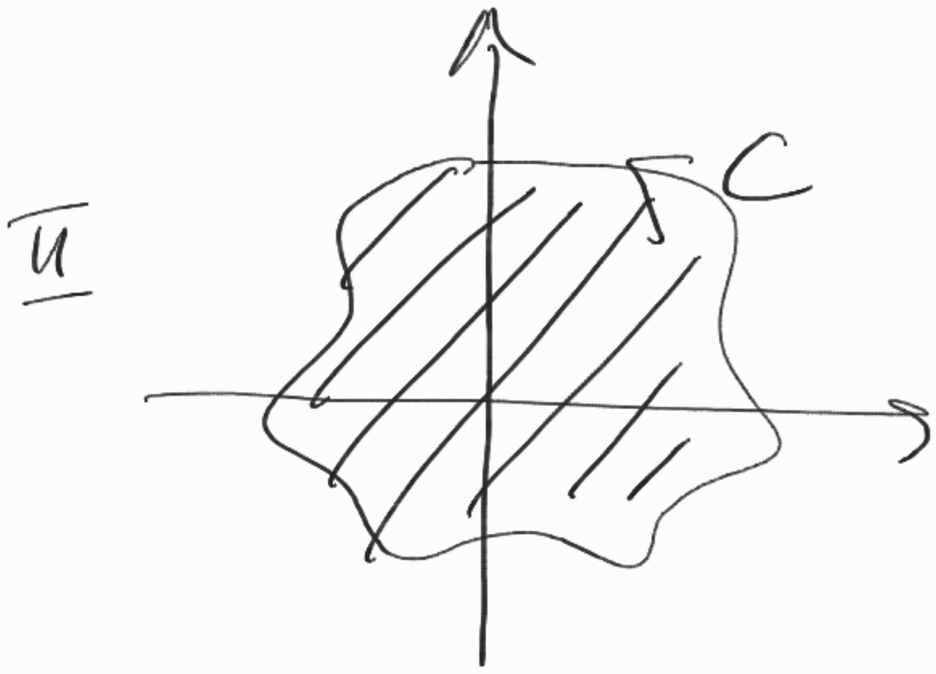


II

Originea nu se află în int (e în ext)

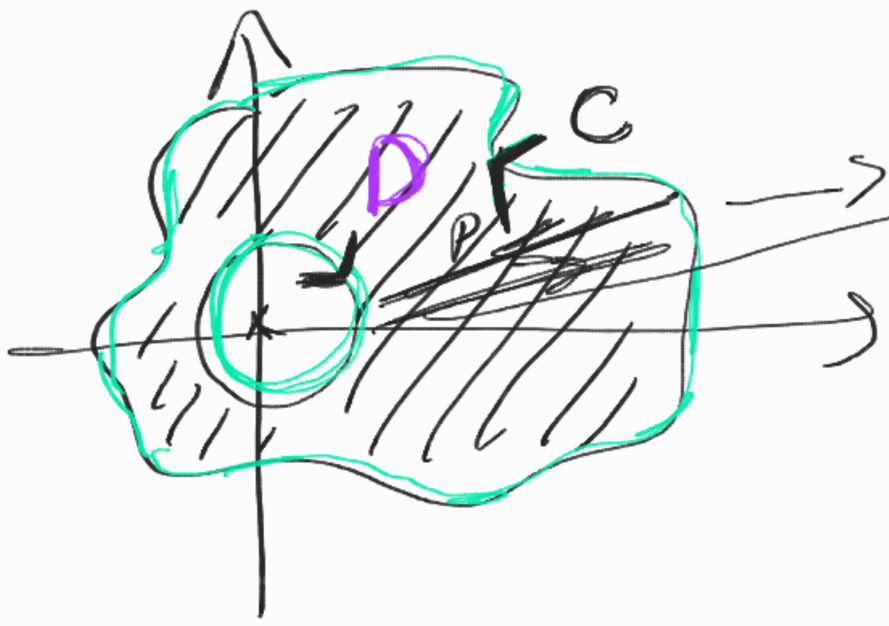
Originea se află în interior

$$\text{I} \quad \int_C w = \iint_D \left(\frac{\partial Q}{\partial x} - \frac{\partial P}{\partial y} \right) dx\,dy =$$

$$= \iint_D \left[\frac{(x^2+y^2) - x \cdot 2x}{(x^2+y^2)^2} + \frac{1 \cdot (x^2+y^2) - y \cdot 2y}{(x^2+y^2)^2} \right] dx\,dy =$$

$$= \iint_D 0 \, dx\,dy = 0$$

II Nu pot aplica Green - Riemann !!!!



frontiera = reuniunea curbei și a cercului

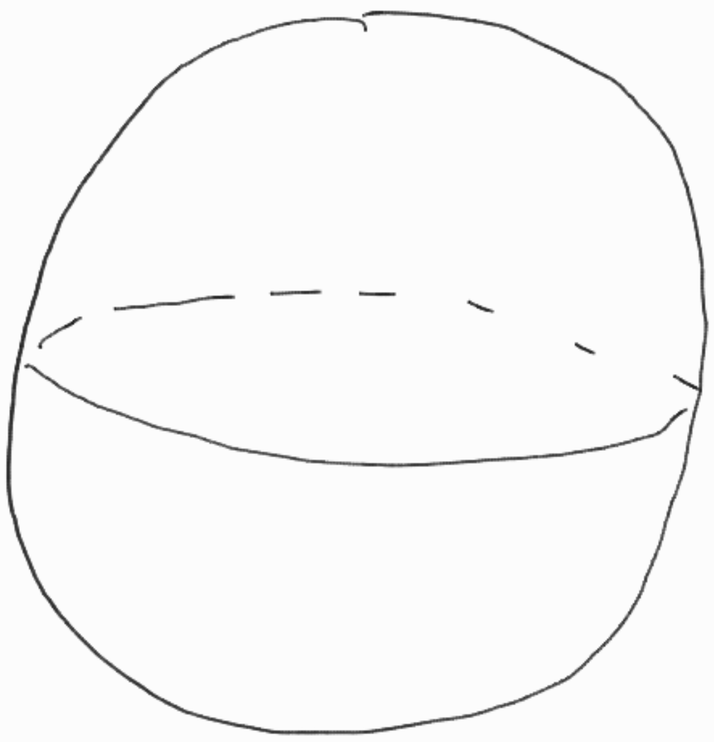
$$\iint_D \left(\frac{\partial Q}{\partial x} - \frac{\partial P}{\partial y}\right) dx\,dy = \int_{C \cup P^- \cup C^- \cup P^+} = \int_C + \int_{C^-} =$$

$$\int_C = 2\pi \Rightarrow \boxed{\int_C = 2\pi}$$

Gauss

Ostrogradski

Formula G-O



$$\bar{F} = P\bar{i} + Q\bar{j} + R\bar{k}$$

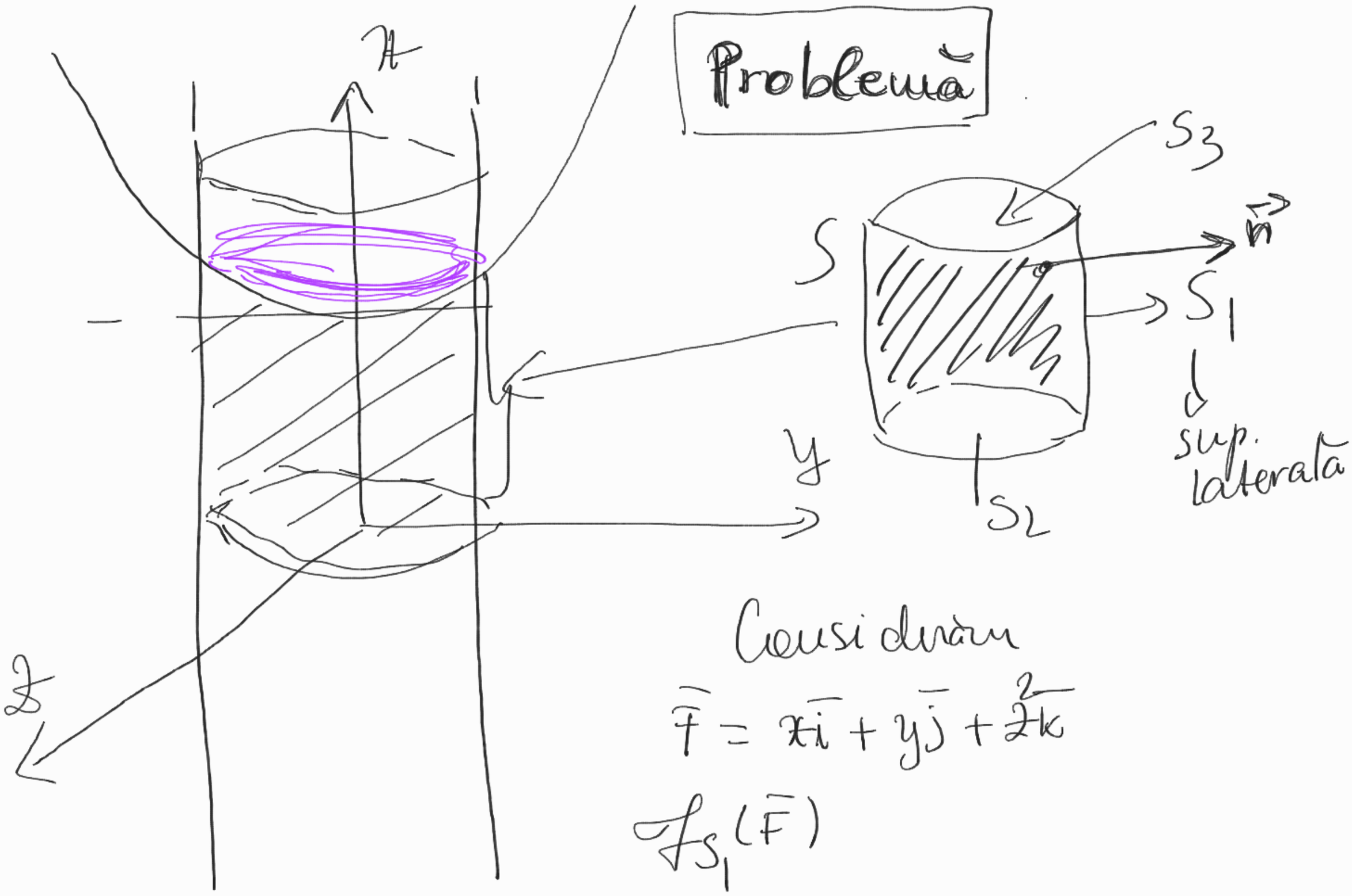
$$\boxed{P, Q, R \in C^1}$$

$$\Phi_S(\bar{F}) = \iint_S \bar{F} \cdot \bar{n}\, d\sigma =$$

$$= \iiint_\Omega \operatorname{div} \bar{F}\, dx\,dy\,dz$$

$$\boxed{\frac{\partial P}{\partial x} + \frac{\partial Q}{\partial y} + \frac{\partial R}{\partial z}}$$

Problema



Consideram

$\bar{f} = x\bar{i} + y\bar{j} + z\bar{k}$

$\int_{S_1}(\bar{F})$

$S_1: \begin{cases} x = \cos t \\ y = \sin t \\ z = u \end{cases} \quad t \in [0, 2\pi] \quad u \in [0,2] \quad \begin{cases} x^2 + y^2 = 1 \\ z = x^2 + y^2 + 1 \end{cases}$

$$\int_{S_1}(\bar{F}) = \iint_D \bar{F} \cdot (\bar{T}_t \times \bar{T}_u)\, dt\, du = *$$

$\bar{T}_t = -\sin t\,\bar{i} + \cos t\,\bar{j} + 0\cdot\bar{k}$

$\bar{T}_u = 0\cdot\bar{i} + 0\cdot\bar{j} + 1\cdot\bar{k}$

$$\bar{T}_t \times \bar{T}_u = \begin{vmatrix} \bar{i} & \bar{j} & \bar{k} \\ -\sin t & \cos t & 0 \\ 0 & 0 & 1 \end{vmatrix} = \cos t\,\bar{i} + \sin t\,\bar{j}$$

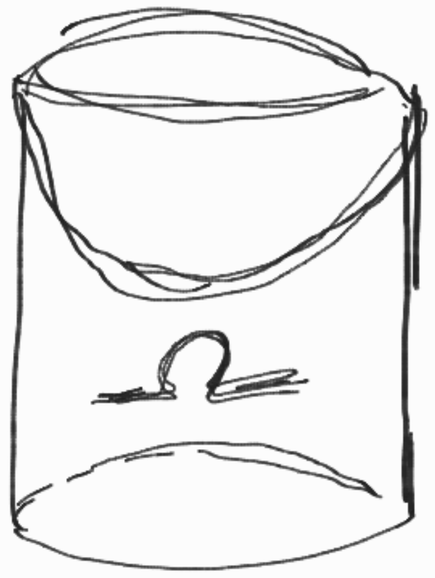
$$* = \int_0^{2\pi}\int_0^2 (\cos t, \sin t, u^2)\cdot(\cos t, \sin t, 0)\, dt\, du =$$

$$= \int_0^{2\pi}\int_0^{2\pi} 1\, dt\, du = 4\pi$$

Pentru flux aplicăm formula G-O.

$$\Phi_S(\bar{F}) = \iiint_\Omega \left(\frac{\partial P}{\partial x} + \frac{\partial Q}{\partial y} + \frac{\partial R}{\partial z}\right) dx\,dy\,dz =$$

$$= \iiint_\Omega (2z+2)\, dx\,dy\,dz \overset{*}{=}$$



$(x,y) \in D$, $x^2+y^2 \le 1$

$z \in [0, x^2+y^2+1]$

$$\overset{*}{=} \iint_\Delta \int_0^{x^2+y^2+1} (2z+2)\, dz\, dx\, dy = \iint_\Delta (z^2+2z)\Big|_0^{x^2+y^2+1} dx\,dy$$

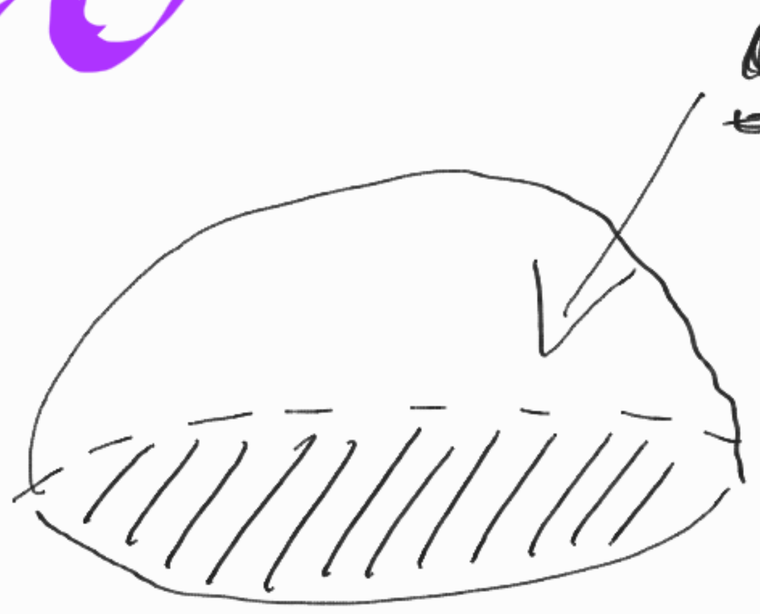
$$= \iint_D (x^2+y^2+1)^2 + 2(x^2+y^2+1)\, dx\,dy = \overset{\text{coordonate polare}}{\cdots}$$

$$= \int_0^{2\pi}\int_0^1 \left[(R^2+1)^2 + 2(R^2+1)\right] \cdot R\, dR\, dt$$

STOKES

$C = \partial S$

b
frontiera
lui S



$$\int_C \bar{F}\, d\bar{r} = \mathcal{C}_C(F) = \mathcal{F}_S(\text{rot}\,\bar{F}) =$$

b circulatia

$$= \iint_S (\text{rot}\,\bar{F}) \cdot \bar{n}\, d\sigma$$

$$\text{rot}\,\bar{F} = \nabla \times \bar{F} = \begin{vmatrix} \bar{i} & \bar{j} & \bar{k} \\ \frac{\partial}{\partial x} & \frac{\partial}{\partial y} & \frac{\partial}{\partial z} \\ P & Q & R \end{vmatrix}$$

Se aplică pe diagonala derivatele

Det simbolic

PROBLEMĂ

cere trigonometrie

$\begin{cases} x = \cos t \\ y = \sin t \end{cases}$

$\uparrow \; z = 8 - \cos^2 t - \sin t$

curbă orientată C

$r(t) = \cos t\, \bar{i} + \sin t\, \bar{j} + (8 - \cos^2 t - \sin t)\bar{k}$

$0 \le t \le 2\pi$

Și câmpul vectorial

$F(x,y,z) = (z^2 - y^2)\bar{i} + (-2xy^2)\bar{j} + (e^{\sqrt{z}}\cos z)\bar{k}$

circulația câmpului F de-a lungul curbei C?

$$\ell_C(\bar{F}) = \int_C \bar{F}\, d\bar{r} \overset{*}{=} \iint_S (\operatorname{rot} F)\cdot \bar{n}\, d\sigma \overset{*}{=}$$

$$\begin{cases} z = 8 - x^2 - y \\ x^2 + y^2 = 1 \end{cases}$$

$$\operatorname{rot} F = \begin{vmatrix} \bar{i} & \bar{j} & \bar{k} \\ \frac{\partial}{\partial x} & \frac{\partial}{\partial y} & \frac{\partial}{\partial z} \\ z^2 - y^2 & -2xy^2 & e^{\sqrt{z}}\cos z \end{vmatrix} = 0\cdot\bar{i} + 2z\bar{j} + (-2y^2 + 2y)\cdot\bar{k}$$

$S: \begin{cases} x = x \\ y = y \\ z = 8 - x^2 - y \end{cases}$

$D: x^2 + y^2 \le 1$

$\overset{*}{=} \iint_D (\operatorname{rot} F)\cdot(\bar{T}_x \times \bar{T}_y)\, dx\, dy$

D

$$\bar{T}_x = \bar{i} + (-2x)\cdot\bar{k}$$
$$\bar{T}_y = \bar{j} - \bar{k}$$

$$\bar{T}_x \times \bar{T}_y = \begin{vmatrix} \bar{i} & \bar{j} & \bar{k} \\ 1 & 0 & -2x \\ 0 & 1 & -1 \end{vmatrix}$$

$$= \boxed{0\cdot\bar{i} + 2x\bar{j} + (-2y^2+2y)\cdot\bar{k}}$$

este orientat în sus

$\Rightarrow$ Stokes:

$$\int_C (\bar{F}) = \Phi_S(\text{rot}\,\bar{F}) =$$

$$= \iint_D (\text{rot}\,\bar{F})\cdot(\bar{T}_x \times \bar{T}_y)\,dx\,dy =$$

$$= \iint_D (0, 16-2x^2-2y, -2y^2+2y)\cdot(2,1,1)\,dx\,dy$$

$$= \iint_D (16-2x^2-2y-2y^2+2y)\,dx\,dy =$$

$$= 2\iint_D (8-x^2-y^2)\,dx\,dy = 2\int_0^{2\pi}\int_0^1 (8-R^2)R\,dR\,dt$$

$x^2+y^2 \le 1$

$x = R\cos t$

$y = R\sin t$